

Perception-Aware Image-Based Visual Servoing of Aggressive Quadrotor UAVs

Chao Qin , Qiuyu Yu , H. S. Helson Go , and Hugh H.-T. Liu , *Member, IEEE*

Abstract—The maintenance of visual features within the sensor field of view poses a significant challenge for underactuated aerial vehicles such as quadrotors, especially during aggressive maneuvers. However, the existing image-based visual servo (IBVS) control methods rely on strict target visibility assumptions or impose excessive constraints on the quadrotor's agility to meet this requirement. Furthermore, the effectiveness of the visibility constraint defined in prior works remains unverified in aggressive flight tests. To address these issues, we present a robust IBVS scheme for quadrotors to perform aggressive maneuvers while ensuring target visibility. Based on the nonlinear model-predictive control framework, we propose a novel underactuation compensation scheme to eliminate the need for a virtual camera frame, which enables us to formulate the true target visibility constraint. We then introduce a quaternion-based representation of spherical visual features to handle the nonsmooth vector field problem on the 2-sphere and derive its corresponding image kinematics. We validate our method through three challenging visual servo tasks where agile maneuvers are desired: fast landing, aggressive long-distance flight, and dynamic object tracking. Extensive simulation and experiment show that our method consistently achieves a target-visible rate of 100% in all the image frames, even under a maximum pitch of 21.04°. The results validate the effectiveness of our visibility constraint under large robot rotations and underscore its importance in enabling robust and aggressive flights.

Index Terms—Image-based visual servoing, nonlinear model-predictive control (NMPC), quadrotor.

I. INTRODUCTION

WITH the recent advancements in vision and navigation technology, unmanned aerial vehicles (UAVs) have become widely utilized in fields, such as videography, delivery,

surveillance, etc. The quadrotor UAV, in particular, has garnered considerable attention from researchers due to its high maneuverability, agile mobility, and vertical take-off and landing capability. The design of the flight controller typically assumes the availability of an onboard position estimator, e.g., the global positioning system (GPS) or the simultaneous localization and mapping (SLAM) system. However, obtaining long-term accuracy in challenging scenarios, such as high-speed indoor flights, could be very difficult. For quadrotors equipped with a camera, visual servoing provides an alternative solution to regulate the quadrotor's pose with respect to a predefined visual target, allowing the navigation of quadrotors in GPS-denied and SLAM-degraded environments.

Generally, visual servo techniques are divided into two categories: image-based visual servo (IBVS) control and position-based visual servo (PBVS) control. PBVS requires explicit reconstruction of the 3-D Cartesian coordinates of the robot or the observed objects [1]. In comparison, IBVS does not need this step, which results in a simpler system structure and more robust performance under camera calibration errors [2].

Despite its advantages, IBVS encounters serious problems in handling aggressive quadrotor maneuvers. The most significant challenge arises from the limited field of view (FOV) of the sensor, which makes target visibility a crucial problem. Unlike PBVS, which has a state estimation layer to predict positional information even when the target is not visible, IBVS will immediately shut down once the target leaves the camera FOV. To the best of our knowledge, none of the existing IBVS literature has demonstrated aggressive quadrotor flight without additional assumptions on target visibility. For instance, Li et al. [3] present a model-predictive-control-based visual servo framework to accomplish aggressive dynamic-obstacle avoidance, and Guo and Leang [4] navigated a high-speed quadrotor through multiple opening with large rotations. Both of them assume that the target feature can stay within the sensor FOV. However, this assumption does not hold in most agile flight scenarios with a single pinhole camera. It is also worth noting that although the significance of target visibility has been acknowledged for decades, many prior works [5], [6], [7] still resort to limiting the quadrotor's maximum velocity, roll, or pitch to ensure target visibility, which cripples the quadrotor's full capacity dramatically. Why is the target visibility problem so tricky for quadrotor UAVs?

The primary reason is that quadrotors are underactuated systems with only four degrees of freedom. They need to rotate to achieve translational motion, but this rotation can also increase

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the image error between the reference and the observed visual features in an unexpected way. To tackle this problem, previous research mainly focused on designing visual features that are invariant to robot rotation. Two popular approaches have emerged: invariant feature approaches and virtual camera approaches. Invariant features are specially designed visual features that exhibit rotation-invariant image kinematics [8]. Fomena et al. [9] utilized distances between the spherical projections of three points as visual features. Guo and Leang [4] proved that invariant features can serve as the flat outputs of a quadcopter system and proposed a geometric tracking controller. However, the use of invariant features induces difficulties in expressing the visibility objective and constraint, leading all the aforementioned methods to take target visibility as a given condition. Virtual camera approaches obtain rotation-invariant image kinematics by transforming the detected features into a virtual image plane. Li et al. [10] present an adaptive backstepping controller to regulate the quadrotor's pose relative to a planar target with arbitrary orientation. Heshmati-alamdari et al. [11] designed a robust nonlinear model-predictive control (NMPC) strategy using image kinematics defined in the virtual camera plane. Zhang et al. [7] took image moments defined in a level virtual image plane as system states to design NMPC. Based on a similar image-moment feature, Lin et al. [12] applied a linear-quadratic regulator for outer-control-loop design, and Zhang et al. [13] incorporated feature path planning to boost control performance. However, a critical limitation of virtual camera approaches lies in the fact that features transformed into a virtual camera frame lose the information on the true camera rotation, leading to inaccurate assessments of the true target visibility. Therefore, even if visibility constraints defined in [7], [14], and [15] are applied, the target visibility requirement still cannot be guaranteed. For example, as shown in Fig. 1(a), although the target object is staying within the virtual camera FOV at all times, it has already left the actual camera FOV after the drone pitching forward, resulting in a mission failure.

The second reason is the high demand for control response. Most previous methods employed an additional cascaded attitude controller to track the computed velocity [16] or acceleration [7] commands, while ignoring the underlying dynamics caused by the inner-loop attitude controller and, hence, leading to the limited response rate. As a result, even if the visibility constraint is adequately addressed, the target may have left the sensor FOV before the quadrotor can respond. To achieve faster response times, it is necessary to produce lower level control commands, such as thrust and angular rates, to eliminate the intermediary attitude controller.

In this article, we present a robust IBVS method for quadrotor UAVs to perform aggressive maneuvers without additional assumptions regarding target visibility. The key idea is to precisely predict the future image coordinate of the actual visual feature within a time window, followed by imposing visibility constraints to ensure that it remains within the specified image areas. To achieve high prediction accuracy, we consider the full dynamic model of the quadrotor, visual feature, and range (i.e., the distance between the feature and the camera). To integrate model prediction, control computation, and constraint enforcement into a unified framework, we adopt NMPC as our control

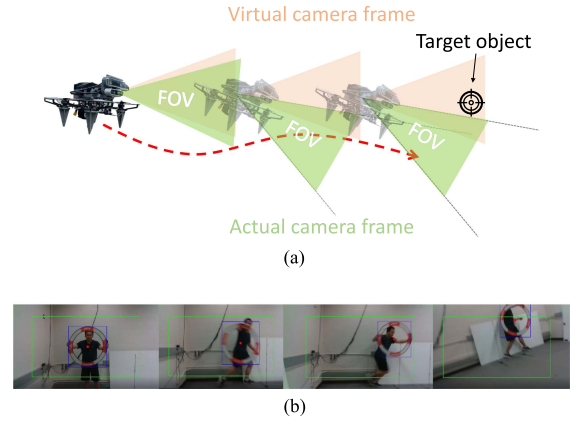


Fig. 1. We aim to formulate true target visibility in IBVS to obtain robust visual servo performance even during aggressive quadrotor maneuvers. (a) Underlying problem of existing virtual camera approaches is that visual features rotated into the virtual camera frame can no longer preserve information about the camera rotation, leading to poor assessment of the true target visibility. As a result, the visibility constraint defined in the virtual camera frame (pink cone) may remain inactive even if the target has left the real camera FOV (green cone). (b) Experimental results demonstrate that our proposed method can maintain the visual feature (the center point of the racing gate) within the specified image area (green rectangle) while performing aggressive maneuvers.

method and directly actuate the quadrotor's thrust and angular rates. The reasons why we prefer NMPC are detailed in the supplementary material [17]. In summary, our contributions are as follows.

- 1) We propose an open-sourced IBVS framework¹ that guarantees target visibility and can be applied to various visual servo tasks requiring aggressive quadrotor maneuvers such as fast landing, aggressive long-distance flight, and dynamic object tracking, as shown in Fig. 1(b), with a video available at <https://www.youtube.com/watch?v=wvtgll-iRx4>.
- 2) A novel underactuation compensation scheme is designed to overcome the aforementioned drawbacks of virtual camera approaches. This scheme enables us to represent the visual feature in the actual image plane such that true target visibility constraints can be well defined.
- 3) We present a novel image kinematic model that respects the manifold structure of spherical image features, leading to a smooth image Jacobian over the full 2-sphere.

II. SPHERICAL IMAGE KINEMATICS

This section introduces the concept of spherical projection, the quaternion representation of a spherical visual feature, and the derivation of the image kinematics.

A. Notation

As shown in Fig. 2, three coordinate frames are used throughout this article: the world frame, $\mathcal{F}^w$, the body frame $\mathcal{F}^b$, and the camera frame, $\mathcal{F}^c$. Let $\mathbf{p}_{lw}^w \in \mathbb{R}^3$ be the global position of a landmark l , which reads “the vector from the origin of frame $\mathcal{F}^w$

¹[Online]. Available: https://github.com/FSC-Lab/fsc_aggressive_ibvs

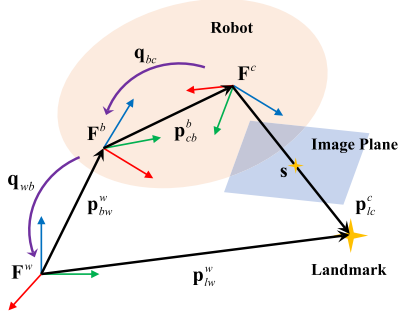


Fig. 2. Schematics representing the world frame $\mathcal{F}^w$, body frame $\mathcal{F}^b$, and camera frame $\mathcal{F}^c$. The standard basis is colored as $\{x, y, z\}$. The relative pose from the camera frame to the body frame is expressed as $(\mathbf{p}_{cb}^b, \mathbf{q}_{bc})$. Similarly, $(\mathbf{p}_{bw}^w, \mathbf{q}_{wb})$ denotes the pose from the body frame to the world frame. A landmark point l located at $\mathbf{p}_{lw}^w$ is projected onto image coordinate at s .

to point l , expressed in frame $\mathcal{F}^w$. A rotation is implemented as a quaternion $\mathbf{q} \in SO(3) \subset \mathbb{R}^4$. The corresponding rotation matrix is denoted as $\mathbf{C}(\mathbf{q}) \in \mathbb{R}^{3 \times 3}$. We use both $\mathbf{q}_{wc}$ and $\mathbf{C}(\mathbf{q}_{wc})$ to represent the rotation from frame $\mathcal{F}^c$ to frame $\mathcal{F}^w$. An operator $\otimes$ is used to denote the Hamilton product of two quaternions. The skew-symmetric matrix of a 3-D vector is written by $(\cdot)^\times$. The 2-D image point expressed in the homogeneous coordinate is denoted by $\mathbf{s} = (u, v, 1)$. Finally, we use $\boldsymbol{\rho} \in S^2 \subset \mathbb{R}^3$ to denote a 3-D unit vector. And the following function is defined to calculate the minimal rotation vector between two 3-D unit vectors, $\boldsymbol{\rho}_1$ and $\boldsymbol{\rho}_2$:

$$\theta(\boldsymbol{\rho}_1, \boldsymbol{\rho}_2) = |\arccos(\boldsymbol{\rho}_1^T \boldsymbol{\rho}_2)| \frac{\boldsymbol{\rho}_1 \times \boldsymbol{\rho}_2}{\|\boldsymbol{\rho}_1 \times \boldsymbol{\rho}_2\|_2}. \quad (1)$$

B. Spherical Image Feature

By projecting the 2-D image coordinate of the visual feature onto the 3-D unit sphere, we can obtain its corresponding spherical image representation. This type of feature resides on the 2-sphere manifold and has been proved to be amenable for underactuated systems [18]. Several methods can be used to parameterize a spherical image feature, including the 2-D minimal representation [2], [14] and the 3-D unit vector representation [6], [18]. However, these methods have distinct limitations that make them less desirable for numerical optimization. The 2-D minimal representation suffers from singularities in the image Jacobian, which can cause numerical issues. On the other hand, the 3-D unit vector representation suffers from the inherent manifold constraint $\|\boldsymbol{\rho}\|_2 = 1$: if simply using three free variables to represent a point on the 2-sphere, the corresponding image Jacobian will be plagued by the “hairy ball theorem,” which states that the image Jacobian function of a 3-D point cannot be smooth over the full 2-sphere, leading to slow convergence rates of NMPC solvers. To overcome this issue, we draw inspiration from [19] and propose a novel feature parameterization for IBVS. This approach addresses all of the aforementioned problems by utilizing a quaternion, $\mathbf{q}$, as the underlying implementation of a spherical image feature.

First, given the target feature $\mathbf{s}$, we utilize the intrinsic parameter matrix $\mathbf{K} \in \mathbb{R}^{3 \times 3}$ of a pinhole camera, as described in [20], to derive the corresponding 3-D unit vector $\boldsymbol{\rho} = \mathbf{K}^{-1}\mathbf{s}/\|\mathbf{K}^{-1}\mathbf{s}\|_2$. Subsequently, we obtain the minimal rotation vector, $\boldsymbol{\alpha} \in \mathbb{R}^3$, between $\boldsymbol{\rho}$ and a constant unit vector $\mathbf{e}_z = [0, 0, 1]^T$, by applying $\boldsymbol{\alpha} = \theta(\boldsymbol{\rho}, \mathbf{e}_z)$. We then convert $\boldsymbol{\alpha}$ into its quaternion representation using the following equation:

$$\mathbf{q} = \begin{bmatrix} \cos \|\boldsymbol{\alpha}\|_2 \\ \frac{\boldsymbol{\alpha}}{\|\boldsymbol{\alpha}\|_2} \sin \|\boldsymbol{\alpha}\|_2 \end{bmatrix}. \quad (2)$$

This resulting quaternion will serve as the spherical image feature for the rest of this article.

It should be noted that the original 3-D unit vector can be recovered simply by applying $\boldsymbol{\rho} = \mathbf{C}(\mathbf{q})\mathbf{e}_z$. And the geometric relationship between $\boldsymbol{\rho}$ and $\mathbf{q}$ can be interpreted as “by rotating $\mathbf{e}_z$ with $\mathbf{q}$, we arrive at a new unit vector that aligns with $\boldsymbol{\rho}$.”

C. Spherical Image Kinematic Model

Given that a rotational quantity is being used to represent a visual feature, it is necessary to use operators on $SO(3)$ to derive its time derivative. To this end, we first define two functions as follows:

$$\mathbf{n}(\mathbf{q}) = \mathbf{C}(\mathbf{q})\mathbf{e}_z \quad (3)$$

$$\mathbf{N}(\mathbf{q}) = [\mathbf{C}(\mathbf{q})\mathbf{e}_x, \mathbf{C}(\mathbf{q})\mathbf{e}_y] \quad (4)$$

where $\mathbf{e}_x = [1, 0, 0]^T$ and $\mathbf{e}_y = [0, 1, 0]^T$. Note that the column space of matrix $\mathbf{N}(\mathbf{q})$ spans the tangent space at $\mathbf{q}$. They can be used to define the following box-plus and box-minus operators on the manifold:

$$\boxplus : SO(3) \times \mathbb{R}^2 \rightarrow SO(3) \quad (5)$$

$$\mathbf{q}, \mathbf{a} \mapsto \exp(\mathbf{N}(\mathbf{q})\mathbf{a}) \otimes \mathbf{q} \quad (6)$$

$$\boxminus : SO(3) \times SO(3) \rightarrow \mathbb{R}^2 \quad (7)$$

$$\mathbf{q}_1, \mathbf{q}_2 \mapsto \mathbf{N}(\mathbf{q}_2)^T \theta(\mathbf{n}(\mathbf{q}_1), \mathbf{n}(\mathbf{q}_2)) \quad (8)$$

where $\exp(\cdot)$ is the exponential map from Lie algebra $\mathfrak{so}(3)$ to Lie group $SO(3)$ [20]. Fig. 3 visualizes the 2-sphere manifold and the geometric relationship among these operators.

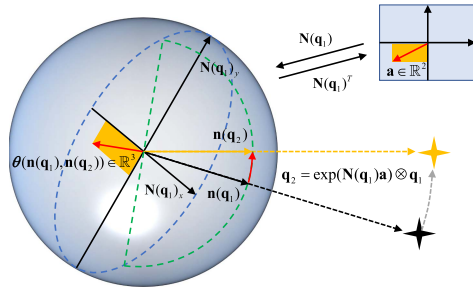
Let us consider a stationary 3-D landmark point l and denote its spherical image feature by $\mathbf{q}$ and its range by r , i.e., the distance from the camera origin to the landmark. We know from frame transformation shown in Fig. 2 that $\mathbf{p}_{lw}^w = \mathbf{C}(\mathbf{q}_{wc})\mathbf{p}_{lc}^c + \mathbf{p}_{cw}^w$. Differentiating this equation with respect to time on both sides, replacing $\mathbf{p}_{lc}^c$ with $\mathbf{n}(\mathbf{q})r$, and applying the chain rule yield

$$\begin{aligned} \mathbf{0} &= \frac{d}{dt}(\mathbf{p}_{lw}^w) = \frac{d}{dt}(\mathbf{C}(\mathbf{q}_{wc})(\mathbf{n}(\mathbf{q})r) + \mathbf{p}_{cw}^w) \\ &= \dot{\mathbf{p}}_{cw}^w + \dot{\mathbf{C}}(\mathbf{q}_{wc})\mathbf{n}(\mathbf{q})r + \mathbf{C}(\mathbf{q}_{wc})\frac{\partial \mathbf{n}(\mathbf{q})}{\partial \mathbf{q}}\dot{\mathbf{q}}r + \mathbf{C}(\mathbf{q}_{wc})\mathbf{n}(\mathbf{q})\dot{r}. \end{aligned} \quad (9)$$

Some of the quantities in (9) are trivial to obtain, such as

$$\dot{\mathbf{p}}_{cw}^w = \mathbf{v}^w \quad (10)$$

$$\dot{\mathbf{C}}(\mathbf{q}_{wc}) = \boldsymbol{\omega}_{cw}^{w \times} \mathbf{C}(\mathbf{q}_{wc}) \quad (11)$$



where $\mathbf{v}^w$ denotes the velocity of the quadrotor expressed in frame $\mathcal{F}^w$ and ω_{cw}^w denotes the angular velocity from frame $\mathcal{F}^w$ to frame $\mathcal{F}^c$, expressed in frame $\mathcal{F}^w$. In contrast, the derivation of $\frac{\partial}{\partial \mathbf{q}} \mathbf{n}(\mathbf{q})$ requires some tricks such as the “small-angle perturbation” and “small-angle approximation” [20]

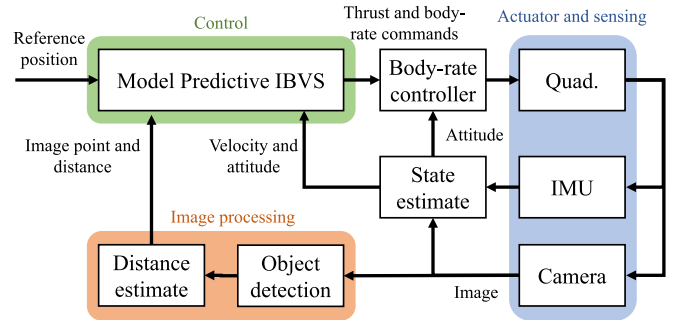
$$\begin{aligned} \frac{\partial}{\partial \mathbf{q}} \mathbf{n}(\mathbf{q}) &= \lim_{\varepsilon \rightarrow 0} \left[\left(\frac{\mathbf{n}(\mathbf{q} \boxplus (e_1 \varepsilon)) - \mathbf{n}(\mathbf{q})}{\varepsilon} \right)^T \right]^T \\ &\approx \lim_{\varepsilon \rightarrow 0} \left[\left(\frac{\mathbf{n}(\mathbf{q} \boxplus (e_2 \varepsilon)) - \mathbf{n}(\mathbf{q})}{\varepsilon} \right)^T \right]^T \\ &= -\mathbf{n}(\mathbf{q})^\times \mathbf{N}(\mathbf{q}) \in \mathbb{R}^{3 \times 2} \end{aligned} \quad (12)$$

According to the IBVS convention, linear and angular velocities expressed in the camera frame are used instead of that expressed in the world frame. Therefore, we use $\mathbf{v}^c = \mathbf{C}(\mathbf{q}_{wc})^T \mathbf{v}^w$ and $\boldsymbol{\omega}^c = \mathbf{C}(\mathbf{q}_{wc})^T \boldsymbol{\omega}_{cw}^w$ to substitute $\mathbf{v}^w$ and $\boldsymbol{\omega}_{cw}^w$ in (9)–(12). Now, we can insert (10)–(12) into (9), followed by left multiplying $\mathbf{C}(\mathbf{q}_{wc})^T$ on both sides to simplify the equation, which yields

$$\mathbf{0} = \mathbf{v}^c - \mathbf{n}(\mathbf{q})^\times \boldsymbol{\omega}^c r - \mathbf{n}(\mathbf{q})^\times \mathbf{N}(\mathbf{q}) \dot{\mathbf{q}} r + \mathbf{n}(\mathbf{q}) \dot{r}. \quad (13)$$

$$\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})^T \left(-\mathbf{n}(\mathbf{q}) \times \frac{\mathbf{v}^c}{r} - \boldsymbol{\omega}^c \right) \quad (14)$$

$$\dot{r} = -\mathbf{n}(\mathbf{q})^T \mathbf{v}^c. \quad (15)$$



A more detailed derivation of the above dynamics model, as well as its corresponding discrete-time kinematic models, which we employ for the prediction of future visual features, is available in our supplementary document [17].

This section introduces the NMPC formulation for the visual servo tasks and our underactuation compensation scheme. The control diagram of the system is illustrated in Fig. 4.

The continuous-time optimal control problem (OCP) can be expressed as the following nonlinear program:

$$\begin{aligned} \min_{\mathbf{x}(t), \mathbf{u}(t)} \quad & \int_{t_0}^{t_f} (\mathcal{L}_a(\mathbf{x}(t), \mathbf{u}(t)) + \mathcal{L}_p(\mathbf{x}(t), \mathbf{u}(t))) dt \\ \text{s.t.} \quad & \dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)) \\ & \mathbf{h}(\mathbf{x}(t), \mathbf{u}(t)) \leq \mathbf{0} \end{aligned} \quad (16)$$

where t_0 is the start time and t_f is the end time. We use $\mathbf{x}$ and $\mathbf{u}$ to represent the system state and control input, respectively. The IBVS system model is denoted as $\mathbf{f}(\mathbf{x}(t), \mathbf{u}(t))$. The objective functions are categorized into two types: the action objective $\mathcal{L}_a(\mathbf{x}(t), \mathbf{u}(t))$ and the perception objective $\mathcal{L}_p(\mathbf{x}(t), \mathbf{u}(t))$. In addition, all the system constraints are encapsulated in the inequality constraint $\mathbf{h}(\mathbf{x}(t), \mathbf{u}(t)) \leq \mathbf{0}$.

The IBVS system model consists of the quadrotor dynamics [21], image kinematics (14), and range kinematics (15). The full state and control inputs are defined as

$$\mathbf{x} = [\mathbf{v}^{w^T}, \mathbf{q}_{wh}^T, \mathbf{q}^T, r]^T \quad (17)$$

$$\mathbf{u} = [c, \boldsymbol{\omega}^{b^T}]^T \quad (18)$$

where $\mathbf{q}_{wb}$ is the orientation of the quadrotor, ω^b is the body rates (a.k.a. ω_{bw}^b), and c is the mass-normalized thrust. Note that this article considers one visual feature, but readers can extend it to multifeature cases. Assume that the camera extrinsic parameters ($\mathbf{q}_{bc}$ and $\mathbf{p}_{ch}^b$) are given. The full dynamic model can be written

as

$$\begin{aligned}\dot{\mathbf{v}}^w &= \mathbf{C}(\mathbf{q}_{wb})\mathbf{c} + \mathbf{g}^w \\ \dot{\mathbf{q}}_{wb} &= \frac{1}{2} \begin{bmatrix} 0 \\ \boldsymbol{\omega}^b \end{bmatrix} \otimes \mathbf{q}_{wb}\end{aligned}\quad (19)$$

$$\begin{aligned}\dot{\mathbf{q}} &= \mathbf{N}(\mathbf{q})^T \left(-\frac{1}{r} \mathbf{n}(\mathbf{q})^\times \mathbf{C}(\mathbf{q}_{bc})^T (\mathbf{C}(\mathbf{q}_{wb})^T \mathbf{v}^w \right. \\ &\quad \left. + \boldsymbol{\omega}^{b^\times} \mathbf{p}_{cb}^b) - \mathbf{C}(\mathbf{q}_{bc})^T \boldsymbol{\omega}^b \right) \\ \dot{r} &= -\mathbf{n}(\mathbf{q})^T \mathbf{C}(\mathbf{q}_{bc})^T (\mathbf{C}(\mathbf{q}_{wb})^T \mathbf{v}^w + \boldsymbol{\omega}^{b^\times} \mathbf{p}_{cb}^b)\end{aligned}\quad (20)$$

where (19) is related to the quadrotor dynamics with $\mathbf{g}^w = (0, 0, -9.81)$ denoting the gravity vector and $\mathbf{c} = (0, 0, c)^T$ the thrust vector; and (20) can be obtained from (14) and (15) by transforming $\mathbf{v}^c$ and $\boldsymbol{\omega}^c$ to $\mathbf{v}^w$ and $\boldsymbol{\omega}^b$, respectively.

It is worth remarking that we consider the range r as a state variable that is propagated in every NMPC iteration, which is a significant deviation from prior approaches [7], [14] that fix the depth to a predetermined value. Incorporating a dynamic range offers several benefits, including enhancing the accuracy of feature coordinate prediction and thereby enabling a more precise assessment of target visibility.

C. Objective Functions

The action objective function aims to regulate the quadrotor's pose with respect to a predefined target object. Let $\mathbf{p}^*$, $\mathbf{v}^*$, $\mathbf{q}_{wb}^*$, and $\mathbf{u}^*$ be the reference position, velocity, orientation, and control inputs, respectively. The objective function can be expressed by

$$\begin{aligned}\mathcal{L}_a(\mathbf{x}(t), \mathbf{u}(t)) &= \|\mathbf{p}^* - P(\mathbf{x}(t))\|_{\mathbf{Q}_p}^2 + \|\mathbf{v}^* - \mathbf{v}^w\|_{\mathbf{Q}_v}^2 \\ &\quad + \|\mathbf{q}_{wb}^* - \mathbf{q}_{wb}\|_{\mathbf{Q}_q}^2 + \|\mathbf{u}^* - \mathbf{u}\|_{\mathbf{R}}^2\end{aligned}\quad (21)$$

where $\mathbf{Q}_p$, $\mathbf{Q}_v$, $\mathbf{Q}_q$, and $\mathbf{R}$ are weight matrices for their corresponding costs. $P(\mathbf{x}(t)) = \mathbf{C}(\mathbf{q}_{wb})(\mathbf{C}(\mathbf{q}_{bc})\mathbf{n}(\mathbf{q})r + \mathbf{p}_{cb}^b)$ compute the rotation-compensated robocentric position of the target feature, which will be further compared with the desired value to obtain the quadratic cost value. This is the essential difference between our work and virtual camera approaches: *we first predict the visual features and then compensate for the robot rotation, while virtual camera approaches do the opposite by first compensating and then predicting*. Furthermore, (21) suggests that our approach utilizes future attitude prediction for compensation, leading to decreased susceptibility to rotation measurement noise.

The perception objective function is designed to align the visual feature with a desired image coordinate $\mathbf{s}^*$, in order to achieve specific perception-related benefits, e.g., orienting the camera towards a target. The function is defined as follows:

$$\mathcal{L}_p(\mathbf{x}(t), \mathbf{u}(t)) = \|\mathbf{s}^* - \pi(\mathbf{q})\|_{\mathbf{Q}_s}^2\quad (22)$$

where $\pi(\mathbf{q}) = \mathbf{K} \cdot \mathbf{n}(\mathbf{q})/\mathbf{n}(\mathbf{q})_z$ maps a quaternion to a 2-D image point. $\mathbf{n}(\mathbf{q})_z$ is the z -axis value and $\mathbf{Q}_s$ is the weight.

It is worth noting that the perception objective can sometimes conflict with the action objective, and as such, its setups can vary depending on the tasks and camera settings. For instance, when tracking a moving object with a forward-facing camera, it is common to assign a higher weight to the horizontal image axis to regulate the yaw angle and maximize target visibility, while assigning zero weight to the vertical image axis to avoid penalizing pitching. However, in a landing task with a downward-facing camera, it may be preferable not to include a perception objective at all, as it could lead to a serious conflict with quadrotor rotations.

D. System Constraints

The system constraints encompass the following inequality constraints:

$$\mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max}\quad (23)$$

$$\mathbf{s}_{\min} \leq \pi(\mathbf{q}) \leq \mathbf{s}_{\max}\quad (24)$$

$$\dot{r}/r \leq \frac{1}{t_{c_{\min}}}\quad (25)$$

where (23) denotes the quadrotor's actuator constraint, (24) the target visibility constraint, and (25) the time-to-collision (TTC) constraint. Specifically, the actuator limit is determined by $(\mathbf{u}_{\min}, \mathbf{u}_{\max})$, while $(\mathbf{s}_{\min}, \mathbf{s}_{\max})$ specifies the image area in which the visual feature should remain. It is worth noting that this area is usually smaller than the full image size. Readers may find the TTC constraint (25) confusing, and we provide a detailed explanation below.

Humans have the ability to perceive TTC t_c , which is the estimated time required to collide with an object if the current velocity is maintained. This information helps in avoiding potential collisions [22] and can be calculated using $t_c = r/\dot{r}$. If t_c is smaller than zero, it means that the range velocity is negative, and the quadrotor can never collide with the target. By maintaining t_c at a value greater than the time threshold $t_{c_{\min}}$, not only can we guarantee a collision-free flight, but we can also avoid large quadrotor rotations caused by immediate braking. Therefore, the safe TTC domain is defined as $\mathcal{T}_c = \{t_c \in \mathbb{R} | t_c < 0 \text{ or } t_c \geq t_{c_{\min}}\}$, which is equivalent to (25). Compared to the pure distance constraint, the TTC constraint provides an additional speed regulation to achieve smooth braking, leading to enhanced target visibility when the target object is close.

E. NMPC Implementation

The system is implemented in C++ and operates in the ROS environment. The NMPC software is based on ACADO, with qpOASES serving as the solver [23]. The prediction horizon is set to 1.0 s with a discretization time step of 0.05 s. Our framework works in a receding-horizon manner by iteratively solving the OCP (16). Regarding real-time performance, the OCP is solved in a time range of 5–10 ms on average, thanks to the real-time iteration (RTI) scheme embedded in ACADO. Readers interested in the technical details of the RTI scheme are referred to [24].

IV. RESULTS

In order to highlight the importance of true visibility constraints to IBVS in aggressive quadrotor flights, three challenging scenarios are considered: fast landing, aggressive long-distance flight, and dynamic object tracking. For the results of normal visual servo tasks where visibility constraints may not be activated, such as hovering and low-speed landing, we put them in our supplementary material [17] along with the comparison with two state-of-the-art methods in [3] and [7]. In the following, our method is mainly compared to the virtual camera approach that also utilizes the target visibility constraint, but in the virtual camera frame, such as in [7]. To ensure a fair comparison, we implement a virtual camera version of our IBVS method based on [7] while keeping the remaining system unchanged.

To quantify the performance of target visibility maintenance, each test will not be terminated when the target is not visible. Instead, we keep providing simulated image measurements, count the number of image frames that cannot detect the visual feature, and define an index called target-visible rate (TVR)—the percentage of the number of image frames that contains the visual feature—to evaluate the effectiveness of the visibility constraint.

A. Fast Landing

While low-speed landings can be completed with near-hover flight, fast landings may require large quadrotor rotations to quickly align with the ground target and descend. However, during large rotations, the visual target may leave the sensor FOV. On the other hand, as the quadrotor descends, the vision will further be limited, leading to a higher risk of losing sight of the target. Therefore, we use this task to validate the effectiveness of the visibility constraint.

We simulate a hummingbird quadrotor with a downward-facing camera using RotorS [25]. The maximum thrust and body rates are 20 N and 3.0 rad/s, respectively. A proportional–integral–derivative controller is developed for angular-rate tracking. In addition, a pinhole camera model with an image size of $752 \times 480 \text{ px}^2$ is used to generate image measurements. We assume that the velocity and orientation of the quadrotor are given. The target of interest is four coplanar points in the world frame: $[-0.4, 0.4, 0.0]^T$, $[-0.4, -0.4, 0.0]^T$, $[0.4, -0.4, 0.0]^T$, and $[0.4, 0.4, 0.0]^T$. When a new image measurement is received, the center of these four points is used as the visual feature and will be constrained within a $451 \times 288 \text{ px}^2$ rectangular image area; the range r is also estimated using the four observed points. The objective is to land the quadrotor from the initial position $[3.0, 3.0, 10.0]^T$ to desired position $[0.0, 0.0, 1.5]^T$ as fast as possible without losing the visual feature in the image plane.

The results of our method are shown in Fig. 5. It can be seen that the quadrotor can successfully land on the desired position with a maximum speed of 4.63 m/s. The maximum roll, pitch, and yaw are 9.30° , 5.88° , and 20.53° , respectively. In Fig. 5(b), we observe that the target feature (yellow dot) departed from image coordinate $[511, 377]^T \text{ px}$, moved to the top right corner of the constrained image area as the quadrotor tilts, and then

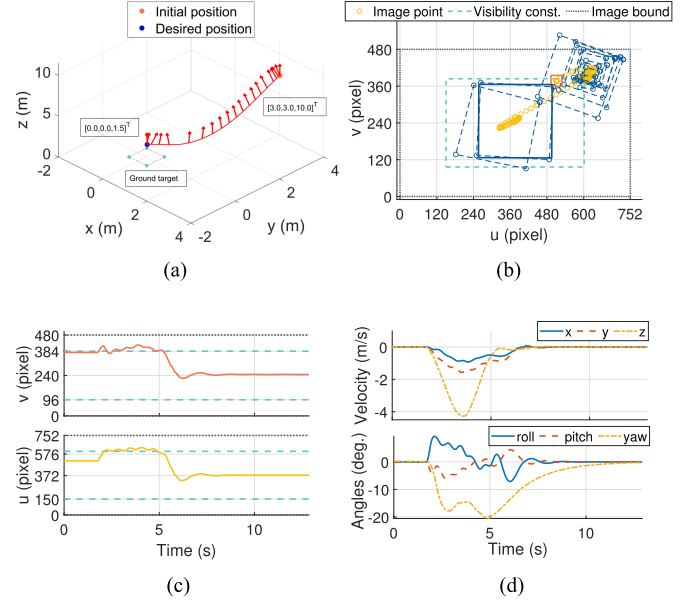


Fig. 5. Trajectories of (a) the quadrotor in the world frame; (b) ground target points (blue dot) with their centers (yellow circle) in the actual image plane (the black dotted-line box denotes the image boundary, the light-blue dashed-line box denotes the specified image area, the red solid-line rectangle denotes the initial feature observation, and the blue solid-line rectangle denotes the terminal feature observation); (c) visual feature coordinate in the image plane; and (d) the velocity and Euler angle of the quadrotor.

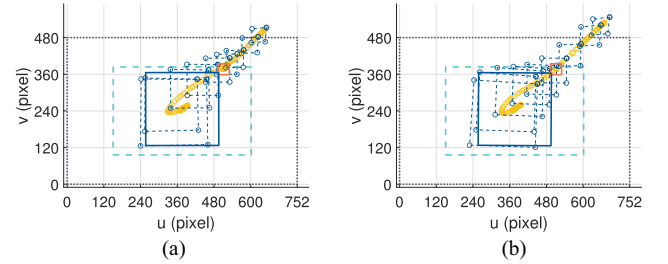


Fig. 6. Visual feature trajectories of (a) our approach without visibility constraint and (b) the virtual camera approach.

came back to the image center as the quadrotor reach the desired position. The trajectory of the visual feature image coordinate is shown in Fig. 5(c). We see that the visibility constraint was active from 1.84 to 5.1 s when the velocity and rotation were large [see Fig. 5(d)]. If the visibility constraint is not applied, which is shown in Fig. 6(a), the target would move outside the image boundary and fail the mission. It suggests that the visibility constraint is effective in enforcing the target feature within the specified area. In addition, by comparing Figs. 5(c) and 6(a), it is clear to see that our method can actively regulate the quadrotor's yaw to ensure target visibility. The results of the virtual camera approach are provided in Fig. 6(b). We see that the visual feature left the image boundary like the unconstrained version of our method shown in Fig. 6(a). It implies that the visibility constraint defined in the virtual camera approach is unable to ensure target visibility.

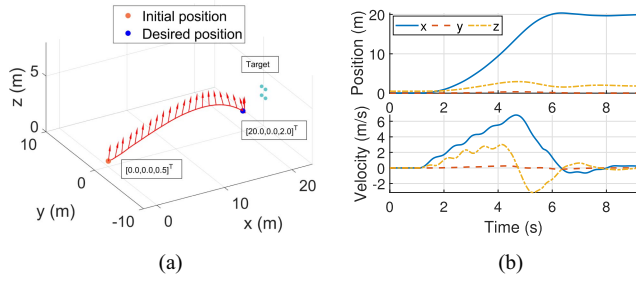


Fig. 7. Trajectories of (a) the quadrotor in the world frame and (b) the position and velocity with respect to time.

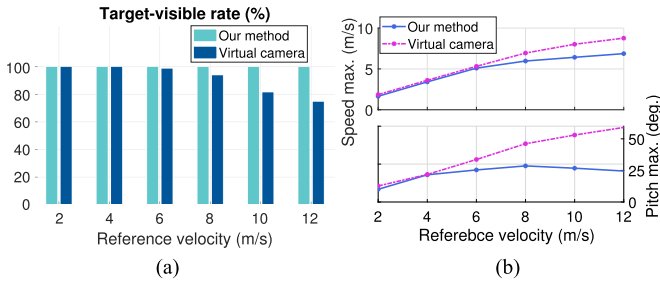


Fig. 8. Comparison of our approach and the virtual camera approach. (a) TVRs under different reference velocities. (b) Curves of the maximum actual speeds and maximum actual pitch angles with respect to the reference velocity.

B. Aggressive Long-Distance Flight

To investigate when and how the visibility constraint affects the control behavior, we test our method in a 20-m flight task where aggressive flight behavior may conflict with the target visibility requirement. For instance, when the quadrotor pitches to gain forward acceleration, the camera also tilts downward, which can result in the potential loss of visual targets. Based on the setup in the last section, we simulate a forward-facing camera on the quadrotor. The target of interest is four coplanar points in the world frame: $[23.0, -0.4, 2.6]^T$, $[23.0, 0.4, 2.6]^T$, $[23.0, -0.4, 3.4]^T$, and $[23.0, 0.4, 3.4]^T$. The objective is to steer the quadrotor from the initial position $[0.0, 0.0, 0.5]^T$ to the desired position $[20.0, 0.0, 2.0]^T$. We tune the desired flight aggressiveness by setting different reference velocities in the action objective function (21).

The results of our method, with a reference velocity of 10 m/s, are displayed in Fig. 7. As expected, the quadrotor successfully reached the desired position with a maximum speed of 6.87 m/s and achieved a 100% TVR even under a maximum pitch of 21.04° . This result indicates that our method can produce aggressive flight while ensuring target visibility. Furthermore, we measured the TVRs of our method and the virtual camera approach under different reference velocities, which are presented in Fig. 8(a). It can be seen that the TVRs of our approach are 100% for all specified reference velocities, whereas the TVR of the virtual camera approach decreases to 74% as the reference velocity increases to 12 m/s, indicating that target visibility is not guaranteed. Meanwhile, from Fig. 8(b), we

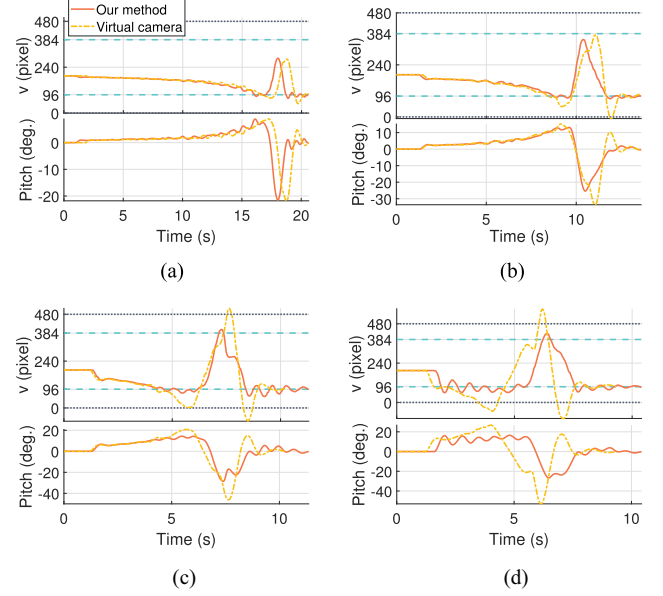


Fig. 9. Trajectories of the vertical image coordinate and the pitch angles under reference velocities of (a) 4, (b) 6, (c) 8, and (d) 10 m/s.

know that our method began to limit the quadrotor's rotation and speed when the reference velocity reaches approximately 6 m/s. These findings demonstrate that our method can prioritize the target visibility requirement and appropriately restrict the quadrotor's agility when desired aggressiveness conflicts with target visibility.

To understand how the visibility constraint works, we gradually increased the desired aggressiveness and observed the resulting change in the visual feature trajectories. Since this task mostly involves quadrotor pitching, we only plot the resulting vertical image coordinate v and pitch angle in Fig. 9, using reference velocities of 4, 6, 8, and 10 m/s. From Fig. 9(a), we know that when the reference velocity is 4 m/s, both our approach and the virtual camera approach obtained similar visual feature trajectories. This is because the visibility constraint has not yet been activated. As we increased the reference velocity to 6 m/s [see Fig. 9(b)], the virtual camera approach started to have features lie outside the image boundary, and it lasted for 0.1318 s. After the velocity grew to 8 and 12 m/s [see Fig. 9(c) and (d)], our approach can still maintain target visibility and give an oscillating feature trajectory around the visibility constraint. And oscillation also occurred in the pitch angle trajectory due to the active visibility constraint. In contrast, the virtual camera approach failed to keep the target within the image boundary and the produced pitch angles were much larger than that of our approach. The result confirms that the visibility constraint defined in our method is capable of regulating the quadrotor's attitude to ensure target visibility, whereas the virtual camera approach cannot.

C. Dynamic Object Tracking

We performed a qualitative evaluation to assess the robustness and real-time performance of our method in a dynamic object

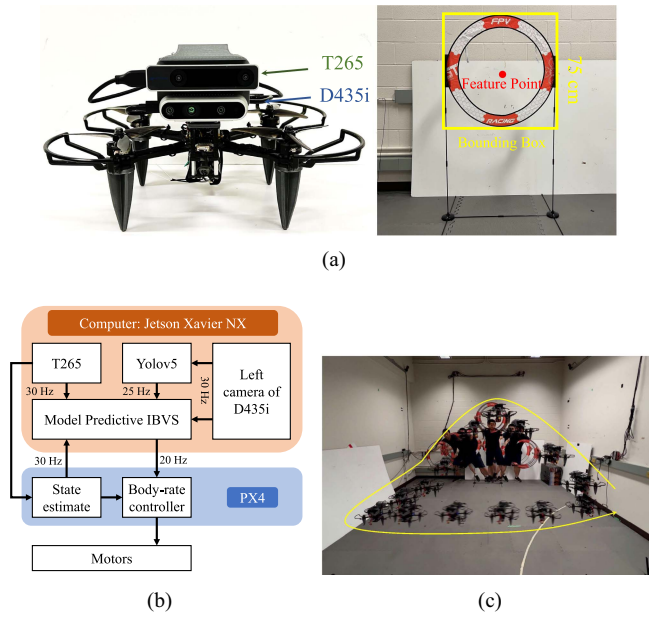


Fig. 10. Experimental system. (a) Hardware for real-world experiment including the first-person-view quadrotor and the visual target, a racing gate. (b) System architecture. (c) Demonstration of dynamic object tracking.

tracking scenario. Specifically, we evaluated whether our approach could successfully track a rapidly moving target, while ensuring that the target feature remained within the specified image area and simultaneously performing aggressive maneuvers. We refer readers to our experimental video.

The experiments are conducted on a quadrotor platform shown in Fig. 10(a) equipped with an Nvidia Xavier NX on-board computer for image processing and visual servo control, a flight controller PixRacer for angular-rate control, and an IntelRealSense camera (only the left one is used with an image size of 1280×640 px²). An additional T265 tracking camera is used to provide velocity estimates. The visual feature is the center point of the racing gate with a diameter of $d_{\text{gate}} = 75$ cm detected by a deep-learning-based object detector, Yolov5. The detector can output a bounding box of the gate, as shown in the yellow box in Fig. 10(a). Using d_{gate} and the size of the detected bounding box, $(w_{\text{box}}, h_{\text{box}})$, we can estimate the range by using $r = \|\rho \cdot d_{\text{gate}} / \max(\frac{w_{\text{box}}}{f_{\text{cam}}}, \frac{h_{\text{box}}}{f_{\text{cam}}})\|_2$, where ρ is the center of the bounding box and f_{cam} is the camera's focal length. The testing environment is a $5 \text{ m} \times 4.3 \text{ m} \times 3 \text{ m}$ indoor space, as shown in Fig. 10(c). The desired range is 2.0 m, and the desired visual feature coordinate is the image center.

The trajectories of the quadrotor in the world frame and the visual feature in the image plane are displayed in Fig. 11(a) and (b), respectively. We see that the tracking was successful and stable. The maximum speed reaches 1.92 m/s, and the maximum roll angle reaches 14.93° . Under such aggressive maneuvers, the visual feature can still be constrained within the specified image area, showing that our approach can support robust and aggressive quadrotor flight.

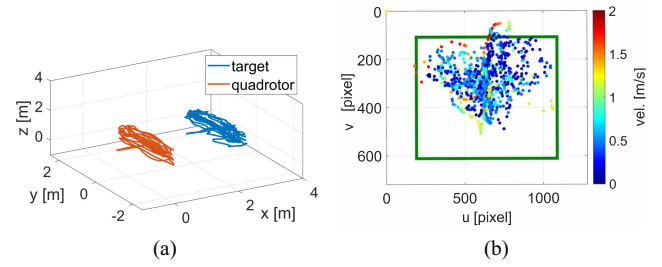


Fig. 11. Experiment results of dynamic object tracking. (a) Trajectories of the quadrotor and the target. (b) Visual features in the image plane (the specified image area is visualized as a green rectangle and the colors of the point indicate the quadrotor's speed when features were captured).

V. CONCLUSION

In this article, we presented a robust IBVS method for aggressive quadrotors without additional assumptions on target visibility. Unlike previous methods that restrict quadrotor agility to achieve target visibility, we achieved this goal by formulating the true target visibility constraint using a novel visual servo mechanism. The effectiveness of our visibility constraint formulation was verified through three challenging visual servo tasks, including fast landing, aggressive long-distance flight, and dynamic object tracking. In all the tasks, our method achieved a 100% target-visible rate while reaching a maximum speed of 6.87 m/s. These results demonstrated that our method can properly restrict quadrotor aggressiveness to maintain target visibility. In future work, we plan to incorporate target dynamics to improve tracking performance and integrate multiple visual features to eliminate the need for yaw estimates.

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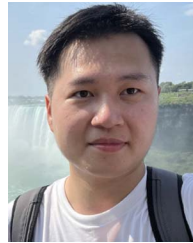
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